

1. RUM - An Overview

- Real User Monitoring is one of the feature provided by Catchpoint.
- Functionality can be utilized by two ways
 - Embedding Catchpoint Javascript
 - Sending details to RealUser server manually in specified format (Containers)

2. Types of RealUser Apps

- RealUser Apps page has got multiple options define apps
 - Site
 - Most commonly used app type for websites
 - Client needs to embed catchpoint javascript
 - Catchpoint JS takes care of populating all the metrics
 - User can monitor using our UI
 - iOS
 - Initially developed to support iOS apps Less or Not used any more
 - Mobile Container
 - Initially Developed to support Mobile Apps less used
 - Site Container
 - Client do not need to embed catchpoint javascript
 - Metrics and dimensions are predefined (All Site App metrics applicable)
 - Client needs to populate metrics and send to our RealUser Server
 - Data should follow specific JSON format
 - Data aggregation with various charts and options can be seen on UI
 - Generic Container
 - Similar to Site Container
 - Unlike Site Container metrics are not predefined
 - User can define either metric(s) or dimension(s) or both
 - Client needs to send data to RealUser Server in JSON format
 - Data aggregation with various charts and options can be seen on UI

3. Testing Considerations for Real User Apps

- a. Creation of Apps
- b. Deletion of Apps
- c. Adding Multiple Site, iOS, Mobile Containers
- d. Site Container Apps and pushing Data
- e. Creating Generic Container and deleting Generic Container

4. Licenses

- License Are Defined at Client Details Page
- Client should have at least one license
- License are specific to App type Example : Site, Mobile, Generic etc.
- License should be added separately for each type

5. Testing Considerations for Real User Licenses

- User should be able to create app only if he has license
- Adding only Generic License should not allow user to create site apps
- Adding only Site App License should not allow user to Create Generic Apps
- User should not be allowed create a app more than defined limit
- Adding Apps should take user to Pop up of selecting RUM license
- User should be restricted to max allowed Page Views Per Day
- User should be restricted to max allowed Storage Days
- User should be restricted to max allowed Page groups
- User should be restricted to max allowed Indicators
- User should be restricted to max allowed Tracepoints
- User should be restricted to max allowed Dataviews
- User should be restricted to max allowed WaterfallSample
- User should be restricted to max allowed Raw Storage Days
- User should be restricted to max allowed Variations
- User should be restricted to max allowed Tracepoints
- User should be restricted to max allowed Zones
- Only Items checked in Store should be shown

6. Dataviews

- Dataviews provide the ability to view pre-aggregated data to make viewing trends over extended time-frames possible.
- There are 2 types of Dataviews:
 - Custom Dataviews: user-defined allowing you to customize the data-set
 - System Dataviews: pre-defined by Catchpoint to get commonly used trends out-of-the-box
 - Site Dataviews are grouped in the following categories: "Content", "Network", "Client", "Tracepoint", "Histogram", and "Engagement".
 - Custom Dataviews can be created under Analysis > Dataviews > Create Dataview

7. Dataviews Testing Considerations

- Considering Level Page, Ajax, Provider And Zones
- Filter should reflect according to Level
- Metrics will be dependent on the Level.
- Metrics shown in the Dataviews should match with Data Explorer
- Adding Fields and Populating Data
- Adding Active and Inactive Apps
- Adding Different Filters
- Adding Filters Blank and Filters with Values
- Creation of Dataview and use it under Different Data Explorer
- Comparing Dataview Data with Partition Report Service and Data Explorer
- Searching Dataviews

- Validating filters in Dataviews In Added Tabs

8. DataExplorer

Explorer allows you to perform multi-dimensional analysis of real user data to quickly discover the causes for performance anomalies by allowing cross-dimensional analysis, such as slowest browsers by city, most visited pages by device, or bounce rate by page by browser. By analyzing data in small periods of time, Data Explorer allows you to analyze in real time the full data set captured. Such functionality and flexibility would not be available in pre-aggregated reports or systems that sample data.

- Data Explorer Provides user to Chart the Real User Data.
- User can Chart the Data by customizing
 - Timeframe
 - Chart Dimensions
 - Breakdown
 - Different Filters
- Data Explorer has below Options
 - Page
 - Ajax
 - Provider
 - Zones

Page

- Page: Data is shown as per the page , Additionally Page provides filter data by Page. Filtering by Page shows the chart for individual page and data for the metrics

Testing Consideration for Page

- When Page is Selected There should be breakdown option Page
- There should be Filter option Page
- Filtering Page should display only for that Page
- Choosing Page Breakdown should display Chart by Page
- There should not be filters or breakdown related to Ajax, Provider, Zones
- Only Metrics related to Pages are shown

Ajax

You can measure the performance of ajax requests made on a page and understand the impact on user experience. You can choose to view the performance of individual ajax requests, or analyze the combined impact that ajax requests are having on page level and user experience metrics. Real User alerts can also be applied to ajax requests to ensure critical requests are performing well.

Testing Consideration for Ajax

- When Ajax is Selected There should be breakdown option Ajax
- There should be Filter option Ajax
- Filtering Page should display only for that Ajax

- Choosing Page Breakdown should display Chart by Ajax
- There should not be filters or breakdown related to Page, Provider, Zones
- There should not be filters or breakdown related to Ajax, Provider, Zones
- Only metrics related to Ajax are shown

Zones

Zones Helps to categorize the Data by Self or Third Party Data. Data is requested from same host then it is categorized as self zone. If it is from external sources then its categorized as Third party

Testing Consideration for Zones

- When Zone is Selected There should be breakdown option Zone
- There should be Filter option Zone
- Filtering Page should display only for that Zone
- Choosing Page Breakdown should display Chart by Zone
- There should not be filters or breakdown related to Ajax, Provider, Zones]
- Only Metrics related to Zones are shown

Provider

Provider helps to categorize adds, widgets from external souces where it is originated from. Provider has two types of data Provider and Provider Type.

Provider is Name of the Provider Serving the request

Provider Type is Nature of business in the Industry it can be Social, Analytics etc..

List of Provider and Provider Types (Refer: <https://support.catchpoint.com/hc/en-us/articles/115000621806>)

Testing Consideration for Providers

- When Zone is Selected There should be breakdown option Zone
- There should be Filter option Zone
- Filtering Page should display only for that Zone
- Choosing Page Breakdown should display Chart by Zone
- There should not be filters or breakdown related to Ajax, Provider, Zones
- Only metrics related to Providers are shown.

Common Filters in DataExplorer

Below are the list of filters that should appear irrespective of Data Exploer (Page/Zone/Provider/Zones)

- Chart Dimension
- Breakdown
- Comparative Time
- Statistical Values
- Variation
- Connection Type
- Time Filter

- Browsers
- Tracepoint Filter
- AS Filter
- Domain
- OS
- Devices
- ISP
- Company
- Screen Resolution
- Location

9. PageGroups And Variations

Page Groups:

For websites that are modular or built on templates, such as ecommerce sites, Catchpoint includes a feature called Page Groups. By indicating that certain pages are all built from the same module or template you can group pages, like product pages, together. Page groups can be defined by using regular expressions to match the URL. For instance, if all product pages contain /products/ in the path, a regex can be used, and Catchpoint will automatically associate the pages with the Page Group requiring no configuration on the site being monitored. Alternatively, you can define Page Groups using tag tokens shown below, which require special tag in each page.

To Configure Page Groups, Navigate to GlimpseAppDetail.aspx, Add Tag Token and URL RegEx

More Detailed information can be found here : <https://support.catchpoint.com/hc/en-us/articles/360002566092#Page>

Variations:

Some websites vary the content of the page delivered to users to test content or features using A/B or multivariate testing. To track the performance of the different tests, Catchpoint has a feature called Variations. Like Page Groups, Variations are determined with tag tokens. They will allow you to track the performance of individual tests running on your pages

Configuration Details can be found here: <https://support.catchpoint.com/hc/en-us/articles/360002566092#Variations>

10. Records and Record Details

Real User (RUM) data can now be seen in the form of waterfall (similar to Synthetic) in order to display request-level detail.

Not all pageviews will have waterfall. Sampling is done across the pageviews where each request can be tracked. Each waterfall is stored in a record, which is a detailed view of a pageview. The RUM

waterfalls can be accessed via the new Records page through Data Explorer and are available in Gold and Platinum licenses.

Please visit Link for Detailed explanation about Records: <https://support.catchpoint.com/hc/en-us/articles/115003583626>

Testing Considerations for Records:

- Verify Adding Filters in Records Page
- Verify Users, Sessions and Records Tabs Data Displayed correctly
- No of records should match with core data
- Histogram by Document Complete (ms) is matched properly
- Are records are displayed as per clients time zones
- Filtering by waterfall should display only records related to waterfall
- Each Records should have View options which should navigate to Record details page
- Records details page should display URL,Page Group,Variation,Time,User Id,Session Id,City,ISP,Bandwidth,Operating System,Browser data correctly
- Records details metrics data should be correct
- User should be able to filter by Host, URL, Provider or any
- Are charts displayed as per the records

11. Estimator

The Estimator allows you to easily correlate performance and user experience metrics with business outcomes via a dual-axis histogram. It can also create "what-if" scenarios based on historical data to create projections, such as "if I reduce page load time by 3 seconds, what will the impact be on conversions?" In addition to projecting change caused by simultaneously varying performance and business metrics, you can also filter to specific page groups, locations, page variations, and custom dimensions.

Real User Estimation visualization can make projections (outputs) using recent RUM data based on goals (inputs) that might be useful from a business perspective.

The Estimation visualization will output data based on the user changing a set of correlated inputs for website type only. For example, you can adjust the number to test the scenario if you improve performance by certain number, revenue will go up to reflect the modification.

Changing certain numbers will define those inputs as Goals. For example, setting a goal to make 1 million in revenue will automatically set revenue as an input. The analysis would then indicate performance, KPI or traffic would need to improve by X number making that a new output. Since traffic is not always easy to influence and control, it should be changed as an input by the model only if there is no other option to satisfy the goal by the user.

Estimation chart has document complete bins as the x-axis. There are dual-y-axes with dropdown menu selection for the metrics listed below.

Goals (Inputs) and Projections (Outputs) metrics:

Performance (Document Complete)

KPI (Conversions, Conversion Rate, Bounces, Bounce Rate, Revenue)

Traffic (Page Views, Visits)

Filters:

Geo (continent/country)

Page Group

Variation

Tracepoint

The chart does not utilize breakdowns/dimensions.

12. Real User Mentoring – Revisited

What is RealUser?

Real user monitoring. Tracks performance from the perspective of user's browser. As opposed to Synthetic monitoring which tracks performance of internet infrastructure (Web, DNS, Traceroute) from nodes running scheduled tests.

Clients embed Glimpse JavaScript on their site and that JavaScript will send performance and Insight data to Orchestra (Core team applications). Once data is processed by Orchestra, we can view it in the UI.

Setting up a RealUser App

Glimpse Licenses define a site's storage limits and features

- Number of page views per day
- Number of days we store data (raw and aggregated)
- # Page Groups, Variations, Indicators, Tracepoints

Sample Rate – Percentage of page views that we capture data for.

Ignore Catchpoint nodes – will not capture data from any browser tests that are run on Catchpoint nodes

Page Groups – allows multiple pages to be viewed as one unit. For example, a company might group all "About" pages together.

Variations – allows a single page to be viewed in multiple ways. For example, a company might have logic on their server to load a page with ads 50% of the time. They can have a variation for when the page has ads and another for the page without ads.

Insight

- Tracepoints – lets you give a particular string value for a tracepoint key. For example, the client can have a tracepoint for which CDN URL a page is using. So they can set the value to be "cdn1.url.com" or "cdn2.url.com" based on some JavaScript logic.
- Indicators – lets you give a particular number value for an indicator key. Indicators can be thought of as custom metrics. Indicators can only be number values.

Alerts – allows the user to define alert conditions for their Glimpse site, similar to Synthetic tests. Alerts are based on timing metrics and can be filtered by time or other filter conditions such as location or Insight values.

Tracking Code – This is the actual code that you embed on your page. We suggest that client's put it at the very top of the page after the opening <head> tag. Some clients do, some use tag managers.

How it works

The tracking code JavaScript gets a timestamp when it loads and allows the user to add Insight metric values (via the RProfiler.addInfo function). It also has code so that we can immediately start capturing JavaScript errors. It then appends an iframe to the page. The iframe loads the main Glimpse JavaScript from our JavaScriptProvider applications. The client can also choose to host our main Glimpse JavaScript themselves by using the `__cpCdnPath` variable.

The request for the main Glimpse JavaScript has the Glimpse site ID. When the JavaScriptProvider receives a request, it looks up the sample rate for the site and includes the sample rate in the JavaScript response.

Main JavaScript gathers performance data using the `performance.timing` API. This API is not available in all browsers (Safari). When the API is not available, we use the timestamps to calculate timing metrics.

On page load, the main JavaScript appends an iframe to the page. The iframe has the URL of the Log Server (a Core application) and all performance and Insight data is passed as a JSON object in the query string. To make sure that the Log Server correctly received the data, the response should have a header named Expires with a value of -1. If data is sent successfully, it can be viewed in the monitoring links (it sometimes takes a few minutes to appear in the monitoring links because Core needs some time to process it).

On page unload, the main JavaScript creates a POST XMLHttpRequest with performance and Insight data in form data. It should have same User ID, Session ID, Site ID, Page ID, and page count as the on load request.

The following table has the keys for the JSON object that we pass to the Log Server. The PostType key (pt) corresponds to the following enumeration:

```
enum PostType {
    OnLoad = 0           // window.onload event
    OnBeforeUnload = 1   // window.onunloadevent
    OnAbort = 2          // triggered if onunload happens before
onload
    Flush = 3            // unused by JS
}
```

Key	Name
v	Version
pt	Post Type
ui	User Id
si	Session Id
di	Site Id
pi	Page view Id
np	Is new page view (1 or 0)
jsc	JavaScript error count
rf	Referrer (page URL)
pc	Page view count
dh	Screen height
dw	Screen width
rc	Redirect count
sv	SPDY version
pr	Is Prerender (1 or 0)
ae	App Errors (Insight – not used)
ind	Indicators
tra	Tracepoints
cv	Is conversion (1 or 0)
rv	Revenue
ri	Revenue Items
pg	Page Group
ab	Variation
dn	DNS
fc	Full Connect
wt	Wait
ld	Load
de	DOM Interactive
dl	DOM Loaded
dc	Document Complete
rp	Response
cl	Content Load
rd	Redirect
sc	Secure Connect
xe	Exit To Entry
el	Entry to On Load

- Yellow indicates user added values
- Green indicates performance metrics

13. Testing Rum with r.p

- Usually developer provides url to test RUM
- As of now <https://portalstage.catchpoint.com/jptester/v3/ajax.aspx> is the link to test
- If you append Glimpse=true it will send water fall data.
<https://portalstage.catchpoint.com/jptester/v3/ajax.aspx?GlimpseWaterfall=true>

To Test

- Navigate to above url in chrome
<https://portalstage.catchpoint.com/jptester/v3/ajax.aspx?GlimpseWaterfall=true>
 - In Developer Option Network tab, Simply refresh the page.
 - Don't forget to check the preserve data check box.
 - We can see r.p request there. Data in this will be sent to server to further processing.
1. Some variables which r.p will have.

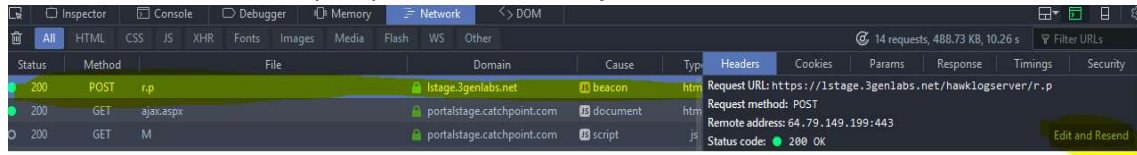
```
"u":"/JSON/dashboard", // URL. No query string or element ID
"pr":0, // protocol. 0 for HTTP. 1 for HTTPS
"pt":2181, // port number
"dn":130, // DNS time
"fc":543, // connect time
"ld":204, // load time
"vct" : //Visually Complete
"wt":5572, // wait time
"dc": //document Complete
"rd":230, // redirect time
"dr":433, // duration time
"ssl":350, // SSL time
"n":100, // count of how many requests were made to URL
"md":0, // method. corresponds to WinHttpMethod enum in
Common.Interfaces
"rp":660, // response time. calculated using ready state changes
"ss":2023, // send size of AJAX request
"rs":96995 // response size of AJAX request
res: is summary data
```

wt: will have data only if it is sending waterfall i.e.
GlimpseWaterfall=true.

ax: key will have only ajax requests made on host, if there is no
ajax request there will not be ax key. On testing url
portalstage.catchpoint.com has ajax request others not so only
portalstage.catchpoint.com should have ax key other should not.

2. r.p Beacon Method

sendBeacon is API and it is another way of sending data back to the server on page unload. So, if check the beacon r.p request content data just click on Edit and Resend



3. Page unload.

Just refresh the page, it will unload the existing page and loads the new page.

For Testing purpose, we have page

<https://portalstage.catchpoint.com/jptester/v3/ajax.aspx>

4. We are not posting r.p data for apps which are inactive

How are we doing it:

Apps which are inactive internally sample rates are set to negative values so data will not be sampled

5. RUM r.p Keys and Description Links

- Excel From Share Point : https://catchpoint-my.sharepoint.com/personal/ghedge_catchpoint_com/Documents/Knowledgebase/RUM%20Specific/JSON%20Schema.xlsx.url?csf=1
- Personal Word Document : https://catchpoint-my.sharepoint.com/personal/ghedge_catchpoint_com/Documents/Knowledgebase/RUM%20Specific/RUM%20key%20values%20for%20r.p%20.docx?d=wc8122a78ffb74bcbfad4cfc632d178a31&csf=1
- OneDrive Excel Sheet: https://catchpoint-my.sharepoint.com/personal/ghedge_catchpoint_com/Documents/Knowledgebase/RUM%20Specific/RUM%20Metrics.xlsx?d=w69186ca2924e4002a54a67153362aa5a&csf=1

6. Partition Report Service Guide for RUM:

- https://catchpoint-my.sharepoint.com/personal/ghedge_catchpoint_com/Documents/Knowledgebase/RUM%20Specific/Partition%20Report%20service%20Querying%20with%20RUM.docx?d=w228bda443e474bc1b95bc4fd8d36272d&csf=1

7. History Page

- History Page is created as a part of SPA (Single Page Application)
- What is Single Page Application in RUM
 - Till date our RUM javascript was capturing static pages or page a whole
 - Now a days Page opens dynamic urls, but page looks as before
 - New views will be loaded in the same page
 - To Track Views in the same page SPA Script helps

- c. As a part of this we added v and nv key
 - d. v key is no of views page
 - e. nv is views loaded and unloaded, Boolean (load =1, unload=0)
 - f. <https://portalstage.catchpoint.com/jptester/v3/History.aspx>
 - Push State : Enter any string will be appened in url consider as new view
 - Back Forward buttons like browser forward and backward button
 - Go text box, go beyond or go forward no. of views (integer)
 - Trigger JS Error Adds JR error into r.p request
 - Make Ajax request Ajax will be loaded into page
 - Execute Script : execute any script
 - g. __cpPreventResourceClear = true : by default it will be false
 - If false : All previous entries in performance metrics will be cleared
 - If true : Preserved
 - The above command should be executed before inline javascript
8. Post any Page Data in <https://portalstage.catchpoint.com/jptester/v3/ajax.aspx>
Simply enter page name/url you want to see in Portal UI and click on post data it will be posted
(Ex: enter www.catchpoint.com click on post data in UI check by breakdown page)
9. Average and SUM metrics
- a. Provider Captures metrics for each pageviews
 - b. Metrics will be summed and averaged for intervals
 - c. Each datapoint is summed average
 - d. Avg metrics, provider keeps aggregating data
 - e. Each datapoint is averaged data
 - f. Finally DataTable for sum metrics will be Average of Sum of Averaged data
 - g. It depends on pageviews and other factors